section*[PZoom Deep Dive 9.5] Verify that α is a transitive set of ordinals

To prove that $\alpha = \{\alpha_x : x \in A\}$ is an ordinal, we show it is a transitive set of ordinals. By construction, every element of α is an ordinal α_x, so it is a set of ordinals.

To prove transitivity, let $\gamma \in \alpha$. This means $\gamma = \alpha_x$ for some specific $x \in A$. We must show $\gamma \subseteq \alpha$. Let $\delta \in \gamma = \alpha_x$.

Under the unique order-isomorphism between the initial segment A_x and the ordinal α_x, the element δ corresponds to some $z \in A_x$, meaning $z < x$. The restriction of this mapping yields an isomorphism between A_z and the initial segment of α_x determined by δ. Because α_x is an ordinal, its initial segment determined by δ is exactly δ itself. Thus, $A_z \cong \delta$.

By the uniqueness of ordinal representatives, because α_z is the unique ordinal isomorphic to A_z, we must have $\delta = \alpha_z$. Since $z \in A$, its representative α_z belongs to α by definition. Therefore, $\delta \in \alpha$.

Since δ was an arbitrary element of γ, this proves $\gamma \subseteq \alpha$. Thus, α is a transitive set of ordinals, which satisfies the conditions of Proposition 2 to be an ordinal itself.

◀ **Return to Main Text**